

Principal Component Analysis and Singular Value Decomposition

1 Unsupervised learning

In some data science applications, the user has access to a set of features (independent variables), $\mathbf{x} = (x_1, \dots, x_p)$ and would like to infer about an output/label y related to the features. So the problem is to learn/find a function f such that $y = f(\mathbf{x})$. Typically one has access to a dataset (training set) to learn about the function f so that given a new input data point $\mathbf{x}^*$, one can predict the output $y^* = f(\mathbf{x}^*)$. Algorithms to find/estimate such functions are often termed *supervised learning*. Since the answer is known (in the training set), the learning stops when an acceptable level of precision in performance is achieved.

In contrast, sometimes we have access to some input data/ features $\mathbf{x} = (x_1, \dots, x_p)$ with no corresponding output. The purpose in such a case is to learn about the structure, distribution, or characteristic of the data. Methods or algorithms to learn in this fashion is often termed as *unsupervised learning*. The goal often is to discover homogeneous subgroups in the data or provide a low-dimensional representation of the data.

Principal component analysis (PCA) is one such useful unsupervised learning technique which helps provide a low-dimensional representation of the data using matrix factorization. In the process we also learn about and another closely-related matrix factorization method, namely Singular Value Decomposition (SVD) as well.

2 Two applications in data analytics

2.1 Image compression

Tool: Singular Value Decomposition (SVD).

The Analytics Edge: Images are an important type of unstructured data on the internet. Companies that share user photos—such as Instagram, Facebook, Flickr, and Twitter—need efficient ways to store them. Data compression is thus an important tool that companies use to tradeoff quality and size of photos. Tools for matrix factorization, such as Singular Value Decomposition (SVD), are particularly useful in this domain. These tools are also adopted in recommendation systems to handle matrices with missing entries.

2.1.1 Overview

Photo and video-sharing social networking services

There are various social networking services that enable users to take and share photos. Instagram is arguably one of the most popular. It was created in 2010 as a free mobile app and acquired by Facebook in 2012. According to Bloomberg, the company is now worth more than \$ 100 billion [?]. Since its launch, more than 40 billion photos have been shared. Another example of such social networking service is Google Photos, a photo and video-sharing platform launched by Google in an attempt to increase the popularity of Google +. So, how many photos are uploaded to the Internet? It is estimated that in 2014 people uploaded an

average of 1.8 billion digital images every single day [?]. Assuming that an image is on average 100 kb, this means that for just one day the amount of bytes needed for storing the images is 1.8×10^{14} bytes, that is, 180 terabytes.

2.1.2 Image representation

A digital (raster) image contains a finite set of pixels, which are arranged in a matrix with a given number of rows and columns. Greyscale images can be seen as matrices in which the value of each pixel (cell) is a single number representing an amount of light—the intensity information. Such images have intensity going from black (0) to white (1), with the different shades of grey taking values in $[0; 1]$. Color images are built as a set of stacked color channels, each of them representing value levels of the given channel. The most common color model is arguably the RGB one, which is based on the three primary colors, red, green, and blue. Hence, an RGB image contains the red, green, and blue color in three separate matrices (generally represented with 8-32 bit).

2.1.3 Summary

Data: Greyscale and color images are stored as matrices containing the intensity information or the information on the red, green, and blue color.

Model: Singular Value Decomposition can be used to extract key information from these matrices.

Value and Decision: an effective image compression system helps store a large amount of images without compromising their quality, and, therefore, the user experience.

2.2 Social Progress Index

Tool: Principal Component Analysis (PCA).

The Analytics Edge: The social progress of nations can be seen as the amalgamation of many features spanning a variety of domains. Providing a measure for social progress can be challenging. We look at a data set of various indicators of social progress and find principal features along which such indicators vary the most with the use of principal component analysis. This aids us in getting a glimpse at how such a social progress index can be computed. We also get to learn one of the useful tools of *unsupervised learning*.

2.2.1 Overview

A recent *New York Times* article discusses the position of the US globally in terms of social progress in light of the 2020 Social Progress Index <https://www.socialprogress.org/>.

The Social Progress Index is a well-established measure, published since 2013, that is meant to catalyze improvement and drive action by presenting social outcome data in a useful and reliable way. Composed of multiple dimensions, the Social Progress Index can be used to benchmark success and provide a holistic, transparent, outcome-based measure of a country's well-being that is independent of economic indicators. Policymakers, businesses, and countries' citizens alike can use it to compare their country against others on different facets of social progress, allowing the identification of specific areas of strength or weakness.

The above is taken from the Methodology summary of the documentation.

2.2.2 Available data

The Social Progress Initiative provides a data set of 192 countries over 10 years (2011-2020) with data collected across 54 different indicators from a variety of sources.

Social Progress Index: These indicators are broken into three broad dimensions:

1. Basic human needs
2. Foundations of wellbeing
3. Opportunity

Each dimension is subdivided into four components further, for example, Basic human needs has components Nutrition and Basic Medical care, Water and Sanitation, Shelter, and Personal Safety. Each component comprises of a weighted sum of a disjoint set of the indicators, the weights are determined using PCA. Now the three dimensions are a simple average of each of their four components; and the average of the three components make the *Social Progress Index*.

The dataset we have also contains all components and dimensions along with the various individual indicators.

2.2.3 Key questions:

1. What can we understand about social progress from the raw data?
2. Can we identify the major differences in social progress occurs?
3. Can we propose some kind of social progress index from our analysis.

2.2.4 Summary

Data: The data contains variables which are linear combinations of others, moreover there are values which are 'not available'. So we need to prepare the data set for analyses.

Model: Principal component analysis helps us in finding key combination of features which explain maximum variability.

Decision: The model provides weights of different indicators without breaking them into components and dimensions. We still get insight into what are the important areas where there is a lot of gap in terms of progress. We can use the key principal component to create a ranking as well.

Value: The value of the model is many-fold. We are able to find the key components of variability allowing a nation to address their social progress issues according to these indicators. Moreover, in a temporal study we may be able to see how different indicators actually become less or more important over time.

3 Matrix factorization

3.1 Spectral decomposition or Eigen decomposition

First let us recall a few things about eigenvalues and eigenvectors of matrices (note that we always consider vectors to be column vectors).

1. Any symmetric matrix $\mathbf{A}$ of size $n \times n$ has orthonormal eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_n$ and associated eigenvalues $\lambda_1, \dots, \lambda_n$ respectively such that:

- (a) $\mathbf{A}\mathbf{u}_j = \lambda_j\mathbf{u}_j$ for all $j = 1, \dots, n$,
- (b) $\mathbf{u}_j^\top \mathbf{u}_j = 1$ for all $j = 1, \dots, n$ (or equivalently $\|\mathbf{u}_j\| = 1$ for all $j = 1, \dots, n$),
- (c) $\mathbf{u}_i^\top \mathbf{u}_j = 0$ for $i \neq j$.

Moreover, $\mathbf{u}_j \in \mathbb{R}^n$ and $\lambda_j \in \mathbb{R}$ for all $j = 1, \dots, n$. By convention, we assume that the eigenvalues are ordered from large to small values:

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n.$$

2. We can express any symmetric matrix $\mathbf{A}$ as:

$$\mathbf{A} = \sum_{j=1}^r \underbrace{\lambda_j \mathbf{u}_j \mathbf{u}_j^\top}_{\text{rank one symmetric matrix}} \quad \text{where the rank of } \mathbf{A} \text{ is } r \leq n.$$

3. A symmetric positive semidefinite matrix has all eigenvalues non-negative.

Note: A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is positive semidefinite if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$. For example, the sample covariance matrix is always positive semidefinite.

3.1.1 Matrix view

For a symmetric matrix $\mathbf{A}$, let $\mathbf{U} = (\mathbf{u}_1 \dots \mathbf{u}_n)$ be the matrix of eigenvectors of $\mathbf{A}$ and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$ where λ_j is the eigenvalue corresponding to $\mathbf{u}_j$. Then $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, namely:

$$\begin{bmatrix} \mathbf{A} \end{bmatrix} = \underbrace{\begin{bmatrix} | \\ \mathbf{u}_j \\ | \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix}}_{\mathbf{\Lambda}} \underbrace{\begin{bmatrix} - & \mathbf{u}_j' & - \end{bmatrix}}_{\mathbf{U}^\top}$$

where $\mathbf{U}\mathbf{U}^\top = \mathbf{I}$ (identity matrix) and $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$.

3.1.2 Geometric view

For a symmetric matrix $\mathbf{A}$, $\mathbf{A}\mathbf{x}$ is a linear transformation of $\mathbf{x}$; stretches the unit ball to an ellipsoid along the direction of the eigenvectors.

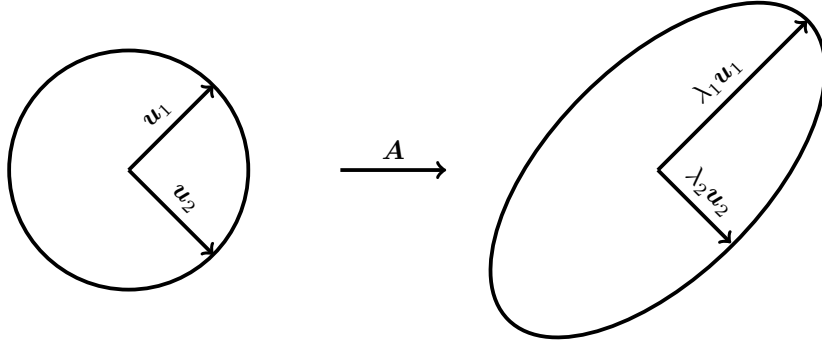


Figure 1.1: Linear transformation: $\mathbf{A}\mathbf{u}_j = \lambda_j\mathbf{u}_j$; stretches the eigenvector by λ_j .

3.1.3 Optimization view

$$\lambda_1 = \max_{\|\mathbf{u}\|=1} \mathbf{u}^\top \mathbf{A} \mathbf{u} \text{ where } \mathbf{u}_1 \text{ is the maximizer.}$$

Using Lagrange multiplier λ , we need to optimize:

$$\mathcal{L}(\mathbf{u}, \lambda) = \mathbf{u}^\top \mathbf{A} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1).$$

Taking partial derivatives, we have

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \lambda} &= -\mathbf{u}^\top \mathbf{u} + 1, \\ \frac{\partial \mathcal{L}}{\partial \mathbf{u}} &= 2\mathbf{A}\mathbf{u} - 2\lambda\mathbf{u}. \end{aligned}$$

Now by equating the above to zero, we get:

$$\mathbf{u}^\top \mathbf{u} = 1, \tag{1.1}$$

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u}. \tag{1.2}$$

Hence (1.1), (1.2) implies that the optimal λ is an eigenvalue of $\mathbf{A}$ and $\mathbf{u}$ is its associated eigenvector. The maximum value of $\mathbf{u}^\top \mathbf{A} \mathbf{u}$ is hence attained for $\mathbf{u} = \mathbf{u}_1$ and the value is λ_1 .

Next, we maximize $\mathbf{u}^\top \mathbf{A} \mathbf{u}$ such that $\mathbf{u}^\top \mathbf{u} = 1$ and $\mathbf{u}$ is orthogonal to $\mathbf{u}_1$. Then:

$$\lambda_2 = \max_{\|\mathbf{u}\|=1, \mathbf{u} \perp \mathbf{u}_1} \mathbf{u}^\top \mathbf{A} \mathbf{u} \text{ where } \mathbf{u}_2 \text{ is the maximizer.}$$

We can continue like this.

3.1.4 Useful fact

For any square matrix $\mathbf{A}$ with eigendecomposition $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, we have

$$\text{trace}(\mathbf{A}) = \sum_{i=1}^n A_{ii} = \sum_{i=1}^n \lambda_i = \text{trace}(\mathbf{\Lambda}).$$

This is true since for matrices $\mathbf{C} \in \mathbb{R}^{m \times n}$, $\mathbf{D} \in \mathbb{R}^{n \times m}$ we have $\text{trace}(\mathbf{CD}) = \text{trace}(\mathbf{DC})$ and hence

$$\text{trace}(\mathbf{A}) = \text{trace}(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top) = \text{trace}(\mathbf{U}^\top \mathbf{U} \mathbf{\Lambda}) = \text{trace}(\mathbf{\Lambda}) = \sum_{i=1}^n \lambda_i.$$

3.2 Singular Value Decomposition

Prior to talking about Principal Component Analysis, we discuss another matrix factorization technique. Any real valued matrix $\mathbf{A}$ of size $m \times n$ (rectangular, not necessarily square) can be expressed as:

$$\mathbf{A} = \sum_{j=1}^r \underbrace{\sigma_j \mathbf{u}_j \mathbf{v}_j^\top}_{\text{rank one matrix}} \quad \text{where the rank of } \mathbf{A} \text{ is } r \leq \min(m, n),$$

and we have

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$$

are the singular values, $\mathbf{u}_1, \dots, \mathbf{u}_r$ are the left singular vectors in $\mathbb{R}^m$ and $\mathbf{v}_1, \dots, \mathbf{v}_r$ are the right singular vectors in $\mathbb{R}^n$. The singular vectors satisfy $\mathbf{u}_j^\top \mathbf{u}_j = 1$ for all $j = 1, \dots, r$, $\mathbf{v}_j^\top \mathbf{v}_j = 1$ for all $j = 1, \dots, r$ and $\mathbf{u}_i^\top \mathbf{u}_j = 0$ for all $i \neq j$ and $\mathbf{v}_i^\top \mathbf{v}_j = 0$ for all $i \neq j$. Here r happens to be the rank of the matrix $\mathbf{A}$.

3.2.1 Matrix view

For a matrix $\mathbf{A}$, we can express the SVD as $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$ where:

$$\underbrace{\begin{bmatrix} \mathbf{A} \end{bmatrix}}_{m \times n} = \underbrace{\begin{bmatrix} \mathbf{U} \end{bmatrix}}_{m \times r} \underbrace{\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}}_{\mathbf{\Sigma} \text{ is } r \times r \text{ matrix}} \underbrace{\begin{bmatrix} \mathbf{V}^\top \end{bmatrix}}_{r \times n}$$

where $\mathbf{U} = (\mathbf{u}_1 \dots \mathbf{u}_r)$, $\mathbf{V} = (\mathbf{v}_1 \dots \mathbf{v}_r)$ and $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_r)$. Here $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ and $\mathbf{U}, \mathbf{V}$ are both semi-unitary, i.e., $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$, $\mathbf{V}^\top \mathbf{V} = \mathbf{I}$.

3.2.2 Geometric view

$\mathbf{A}\mathbf{x}$ is a linear transformation of $\mathbf{x}$; it *rotates* and stretches the unit ball to an ellipsoid.

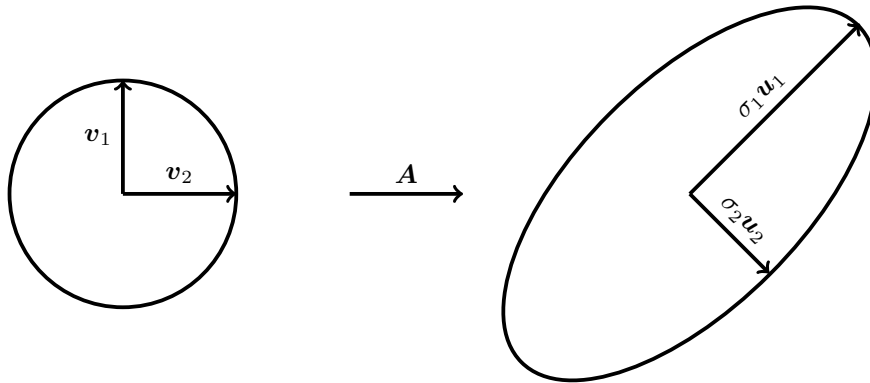


Figure 1.2: Linear transformation: $\mathbf{A}\mathbf{v}_j = \sigma_j \mathbf{u}_j$; it rotates and then stretches the right singular vector.

3.2.3 Relationship between eigendecomposition and singular value decomposition

Consider the matrix $\mathbf{A}\mathbf{A}^\top$:

$$\begin{aligned}\mathbf{A}\mathbf{A}^\top &= (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top) (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top \\ &= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \mathbf{V}\mathbf{\Sigma}^\top \mathbf{U}^\top \\ &= \mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^\top \mathbf{U}^\top \quad (\text{since } \mathbf{V}^\top \mathbf{V} = \mathbf{I}) \\ &= \mathbf{U} \begin{bmatrix} \sigma_1^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_r^2 \end{bmatrix} \mathbf{U}^\top.\end{aligned}$$

The column vectors in $\mathbf{U}$ are exactly the eigenvectors of $\mathbf{A}\mathbf{A}^\top$ (symmetric positive semidefinite matrix) and $\sigma_1^2, \sigma_2^2, \dots$ are the positive eigenvalues of $\mathbf{A}\mathbf{A}^\top$.

Consider the matrix $\mathbf{A}^\top \mathbf{A}$:

$$\begin{aligned}\mathbf{A}^\top \mathbf{A} &= (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top) \\ &= \mathbf{V}\mathbf{\Sigma}^\top \mathbf{U}^\top \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \\ &= \mathbf{V}\mathbf{\Sigma}^\top \mathbf{\Sigma}\mathbf{V}^\top \quad (\text{since } \mathbf{U}^\top \mathbf{U} = \mathbf{I}) \\ &= \mathbf{V} \begin{bmatrix} \sigma_1^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_r^2 \end{bmatrix} \mathbf{V}^\top.\end{aligned}$$

The column vectors in $\mathbf{V}$ are exactly the eigenvectors of $\mathbf{A}^\top \mathbf{A}$ (symmetric) corresponding to $\sigma_1^2, \sigma_2^2, \dots$ which are the positive eigenvalues of $\mathbf{A}^\top \mathbf{A}$.

3.2.4 Optimization view

$$\sigma_1 = \max_{\|\mathbf{v}\|=1} \|\mathbf{A}\mathbf{v}\| \text{ where } \mathbf{v}_1 \text{ is the maximizer.}$$

Using Lagrange multiplier λ , we need to optimize:

$$\mathcal{L}(\mathbf{v}, \lambda) = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A} \mathbf{v} - \lambda(\mathbf{v}^\top \mathbf{v} - 1)$$

Taking partial derivatives, we have

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \lambda} &= -\mathbf{v}^\top \mathbf{v} + 1, \\ \frac{\partial \mathcal{L}}{\partial \mathbf{u}} &= 2\mathbf{A}^\top \mathbf{A} \mathbf{v} - 2\lambda \mathbf{v}.\end{aligned}$$

Now by equating the above to zero we get

$$\mathbf{v}^\top \mathbf{v} = 1, \tag{1.3}$$

$$\mathbf{A}^\top \mathbf{A} \mathbf{v} = \lambda \mathbf{v}. \tag{1.4}$$

Hence (1.3), (1.4) implies that the optimal λ is the largest eigenvalue of $\mathbf{A}^\top \mathbf{A}$ and $\mathbf{v}$ is its associated eigenvector. Therefore the optimal value of $\lambda = \sigma_1^2$ and the corresponding maximizer is $\mathbf{v}_1$, the right singular value (see Section 3.2.3 for an explanation).

Next, we may want to maximize $\|\mathbf{A}\mathbf{v}\|$ such that $\mathbf{v}^\top \mathbf{v} = 1$ and $\mathbf{v}$ is orthogonal to $\mathbf{v}_1$. Then:

$$\sigma_2 = \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1} \|\mathbf{A}\mathbf{v}\| \text{ where } \mathbf{v}_2 \text{ is the maximizer.}$$

We can continue like this.

3.2.5 Low Rank Approximation

The main idea is to develop a low rank approximation (“simple” version) of $\mathbf{A}$ by dropping the smaller singular values and corresponding singular vectors. A rank k approximation $\hat{\mathbf{A}}$ of $\mathbf{A}$ where $k \leq r$ (here r is the rank of $\mathbf{A}$) is given by:

$$\hat{\mathbf{A}} = \sum_{j=1}^k \sigma_j \mathbf{u}_j \mathbf{v}_j^\top \quad (1.5)$$

where $\sigma_1 \geq \dots \geq \sigma_k > 0$ are the k largest singular values of $\mathbf{A}$. Here:

$$\underbrace{\begin{bmatrix} \hat{\mathbf{A}} \end{bmatrix}}_{m \times n} = \underbrace{\begin{bmatrix} \hat{\mathbf{U}} \end{bmatrix}}_{m \times k} \underbrace{\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_k \end{bmatrix}}_{\hat{\Sigma} \text{ is } k \times k \text{ matrix}} \underbrace{\begin{bmatrix} \hat{\mathbf{V}}^\top \end{bmatrix}}_{k \times n}$$

Here we use only the first k columns of $\mathbf{U}$ and first k columns of $\mathbf{V}$ to create $\mathbf{V}^\top$.

Why are “good” low rank approximations popular?

1. To store a full $m \times n$ matrix, we need to store mn values. To store a rank k approximation, we need to store $mk + nk + k$ values which can be much smaller if k is small compared to m and n .
2. In some cases, we believe that a low rank approximation is a natural representation of reality. For example in a ratings matrix where the rows denotes users, columns denote movies and the entries are ratings given by users to movies, we might want to use a low rank approximation. Here the low rank approximation captures the belief that only a few factors might be influencing the users rating of movies.

To understand the quality of a low rank approximation, we need the definition of the Frobenius norm of a matrix. The Frobenius norm of a $m \times n$ matrix $\mathbf{A}$ is defined by:

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2}.$$

Equivalently, the Frobenius norm is given by:

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2} \text{ where } r \text{ is the rank of the matrix}$$

and $\sigma_1 \geq \sigma_2 \geq \dots \sigma_r > 0$ are the positive singular values of $\mathbf{A}$. This is true since for matrices $\mathbf{C} \in \mathbb{R}^{m \times n}$, $\mathbf{D} \in \mathbb{R}^{n \times m}$ we have $\text{trace}(\mathbf{CD}) = \text{trace}(\mathbf{DC})$ and hence

$$\begin{aligned}
 \|\mathbf{A}\|_F^2 &= \sum_{i=1}^m \sum_{j=1}^n A_{ij}^2 = \text{trace}(\mathbf{AA}^\top) \\
 &= \text{trace}\left(\left(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top\right)\left(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top\right)^\top\right) \\
 &= \text{trace}(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top\mathbf{V}\mathbf{\Sigma}^\top\mathbf{U}^\top) \\
 &= \text{trace}(\mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^\top\mathbf{U}^\top) \\
 &= \text{trace}(\mathbf{U}^\top\mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^\top) \\
 &= \text{trace}(\mathbf{\Sigma}\mathbf{\Sigma}^\top) = \sum_{i=1}^r \sigma_i^2.
 \end{aligned}$$

It turns out that the best approximation of a matrix $\mathbf{A}$ with rank r , by any rank $k \leq r$ matrix where the error is measured in the Frobenius norm is exactly given by the formula in (1.6). Namely the optimal solution of:

$$\min_{\text{rank}(\hat{\mathbf{A}}) \leq k} \|\mathbf{A} - \hat{\mathbf{A}}\|^2$$

is given by

$$\hat{\mathbf{A}} = \sum_{j=1}^k \sigma_j \mathbf{u}_j \mathbf{v}_j^\top \quad (1.6)$$

where $\sigma_1 \geq \dots \geq \sigma_k > 0$ are the k largest singular values of $\mathbf{A}$ (with $k < r$) and the optimal objective value for (1.6) is $\sigma_{k+1}^2 + \dots + \sigma_r^2$. When approximating a matrix by a low rank matrix, the accuracy of the approximation is given by:

$$\frac{\|\hat{\mathbf{A}}\|_F^2}{\|\mathbf{A}\|_F^2} = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^r \sigma_i^2}$$

We will see similar quantities appearing for PCA as well.

3.3 Principal Component Analysis

Problem set up:

1. n = number of observations
2. p = number of features
3. $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^\top$ = i -th observation with p feature variables (for variable $\mathbf{X} = (X_1, \dots, X_p)^\top$)
4. We observe the $n \times p$ data matrix:

$$\mathbb{X} = \begin{bmatrix} x_{11} & \dots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix}.$$

5. We first center the data so that it has sample mean zero for each feature:

$$\sum_{i=1}^n x_{ij} = 0 \text{ for all } j = 1, \dots, p.$$

6. Denote by $\mathbf{S}$, the sample covariance matrix of $\mathbf{x}$ given by

$$\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top = \frac{1}{n-1} \mathbb{X}^\top \mathbb{X}$$

where $\mathbf{S}$ is a positive semidefinite symmetric square matrix of size $p \times p$ (unbiased estimator of the true covariance matrix $\mathbf{Q}$).

Question: How do we find a low-dimensional representation of the data? More formally, find a projection of the data that best explains the variation in the data.

Example: Let's look at the following example (plots). The data in 1.3 was generated from the normal distribution where:

$$\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{Q} = \begin{bmatrix} 5 & 5 \\ 5 & 10 \end{bmatrix}; \text{ with } \lambda_1 = 13.09017, \mathbf{v}_1 = \begin{bmatrix} 0.5257311 \\ 0.8506508 \end{bmatrix}, \lambda_2 = 1.90983, \mathbf{v}_2 = \begin{bmatrix} -0.8506508 \\ 0.5257311 \end{bmatrix}$$

As we can see from the figure, the eigenvector $\mathbf{v}_1$ of the covariance matrix $\mathbf{Q}$ is the first principal direction which is exactly the direction that explains most of the variability in the data. The next direction orthogonal to $\mathbf{v}_1$ which explains most of the variability in the data is $\mathbf{v}_2$.

3.3.1 Finding low-rank representations: PCA

To formalize this, observe that we want to find an unit vector $\mathbf{v}$ of unit norm (direction) such that:

$$\max_{\|\mathbf{v}\|=1} \text{Variance}(\mathbf{X}^\top \mathbf{v})$$

where $\mathbf{X} = (X_1, \dots, X_p)^\top$ is a random vector with zero mean and covariance matrix $\mathbf{Q}$. Recall that a projection of a vector $\mathbf{X}$ on a vector $\mathbf{v}$ of unit norm is given by the vector:

$$(\mathbf{X}^\top \mathbf{v}) \mathbf{v}.$$

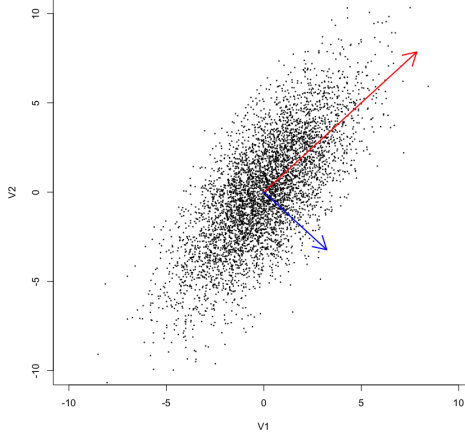


Figure 1.3: 5000 sample data points from a normal distribution in $\mathbb{R}^2$. Eigendecomposition of the covariance matrix identifies eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ and eigenvalues λ_1, λ_2 . The red arrow shows $\lambda_1 \mathbf{v}_1$ and the blue arrow shows $\lambda_2 \mathbf{v}_2$ (upto a scaling constant).

Since $\mathbf{X}$ is random we need to find this on an average and can express the optimization problem as:

$$\max_{\|\mathbf{v}\|=1} \mathbb{E}[(\mathbf{X}^\top \mathbf{v})^2] = \max_{\|\mathbf{v}\|=1} \mathbf{v}^\top \mathbb{E}[\mathbf{X} \mathbf{X}^\top] \mathbf{v} = \max_{\|\mathbf{v}\|=1} \mathbf{v}^\top \mathbf{Q} \mathbf{v}.$$

The solution to this as we saw before is exactly the first eigenvector of $\mathbf{Q}$. The direction $\mathbf{v}_2$ is the next principal direction which best explains the variation in the dataset orthogonal to $\mathbf{v}_1$ and so on. Since we do not see the true covariance matrix $\mathbf{Q}$ of the population, we use the sample covariance matrix $\mathbf{S}$ while doing PCA and compute the eigendecomposition of $\mathbf{S} = \frac{1}{n-1} \mathbb{X}^\top \mathbb{X}$ or equivalently compute the SVD of $\frac{1}{\sqrt{n-1}} \mathbb{X}$ (the eigenvectors of $\frac{1}{n-1} \mathbb{X}^\top \mathbb{X}$ are simply the right singular vectors of $\frac{1}{\sqrt{n-1}} \mathbb{X}$). Keep in mind the difference between random variable $\mathbf{X}$ and data matrix $\mathbb{X}$.

In principal component analysis, we compute the first few principal components and then project the data onto the subspace spanned by them to reduce the dimension of the data. The projection of a vector $\mathbf{x}$ on the subspace spanned by these eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ which are of unit norm and orthogonal to each other is given by the vector:

$$(\mathbf{x}^\top \mathbf{v}_1) \mathbf{v}_1 + \dots + (\mathbf{x}^\top \mathbf{v}_k) \mathbf{v}_k.$$

Now the total variance in the data set is:

$$\begin{aligned} \text{Total (sample) variance} &= \text{Variance}(X_1) + \dots + \text{Variance}(X_p) \\ &= \frac{1}{n-1} \sum_{j=1}^p \sum_{i=1}^n x_{ij}^2 = \text{Trace} \left(\frac{1}{n-1} \mathbb{X}^\top \mathbb{X} \right) \\ &= \text{Trace}(\mathbf{S}) \\ &= \text{Trace}(\mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top) \\ &= \text{Trace}(\mathbf{V}^\top \mathbf{V} \mathbf{\Lambda}) = \text{Trace}(\mathbf{\Lambda}) = \sum_{i=1}^p \lambda_i \end{aligned}$$

where λ_i are the eigenvalues of $\mathbf{S}$. When choosing the first k principal components of $\mathbf{S}$, we define a quantity R^2 which explains the proportion of variance explained by these components:

$$R^2 = \frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^p \lambda_i}$$

where λ_i are the eigenvalues of $\mathbf{S}$ sorted from highest to lowest.

Depending on a threshold chosen by the user, the number of principal components can be determined. In practice this is done using a scree plot.

In practice we observe that the different attributes have different variances and hence a PCA gets influenced by the variable with largest variance. Hence it is customary to standardize all variables to have unit (or equal variance). If all variances are equal then the total variance is equal to $\sum^p \lambda_i = p$. In such a case, it is often recommended that the principal components are chosen for eigenvalues with value greater than 1, since they explain more than one variable of the entire set.

3.3.2 Principal Component Analysis in R

1. Two functions are generally used `prcomp` and `princomp`. We use the first one.
2. The output, say `pr.out <- prcomp(data)` stores a few values
 - `pr.out$center` and `pr.out$scale` are means and standard deviations of the data which are used to standardize the data.
 - `pr.out$sdev` are standard deviations of the principal components (square root of the eigen values of the $\mathbf{S}$ matrix).
 - `pr.out$rotation` are the attribute loadings (eigen-vectors) and `pr.out$x` are the individual scores according to the principal components.
3. A Scree plot can be useful and `plot(pr.out)` creates such a plot.
4. We tend to plot also the cumulative proportion of variance explained by the principal components as discussed in 3.3.1.
5. We use `library(factoextra)` to do some colorful plots.

3.3.3 Limitations of PCA

Albeit being quite useful, we need to keep in mind certain limitations of PCA.

- PCA depends on the scaling of the variables and hence one needs to scale the variables, essentially bringing them all down to the same scale. For example sometimes different features (attributes) have very different variances and hence a PCA gets influenced by the variable with largest variance. It is customary to standardize all variables to have unit (or equal variance).
- The eventually identified principal components might lose some interpretability.
- All of this works under the assumption that a linear relationship between variables can explain the total variability.